The Sound of Power: An NLP Analysis of United Nations Security Council Speeches on Crisis Management

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Abstract

This study examines how member states of the United Nations Security Council (UNSC) use language to justify their positions on crisis and conflict management, specifically focusing on debates surrounding sanctions, humanitarian aid, and ceasefires. While existing literature has predominantly analyzed *what* topics emerge in UNSC discourse and *who* introduces new agenda items, our research investigates *how* countries strategically use rhetoric to argue their positions. We apply a multi-method computational framework, comprising Stance Detection, Rhetorical Strategy Analysis, Topic Modeling (LDA and BERTopic), Framing Analysis, and Transformer-based embeddings, to extract over 450 linguistic features per speech. Our analysis reveals that linguistic variability in pre-vote speeches is driven primarily by resolution type rather than country identity or bloc membership. Speeches cluster around meeting topics, indicating that UNSC diplomatic discourse is highly regimented and formulaic, shaped by institutional conventions regardless of geopolitical orientation. Rather than treating ambiguity and hedging as analytical noise, we argue these features constitute essential mechanisms for negotiating power and legitimacy within the Council. This work demonstrates the importance of analyzing how states argue, not merely what they say, providing a framework for examining how institutional context shapes diplomatic language in multilateral decision-making.

1 Introduction

This research addresses a critical gap in computational political science by examining whether diplomatic rhetoric can be quantified to reveal national communication styles and power hierarchies within the UNSC. While UN resolutions have cascading impacts on global security, trade, and human rights, understanding how member states justify their positions through language offers insights into underlying mechanics of discursive power and the rhetorical identities that shape multilateral diplomacy (Eckhard et al., 2023). To find this, we examine whether observable linguistic patterns can reveal underlying geopolitical alignments and power hierarchies that manifest in rhetorical strategy choices. Prior works have focused on mapping policy preferences through text (Laver & Benoit, 2018), but this work treats the language itself, specifically, its ambiguity, hedging, and framing, as the primary phenomenon of interest rather than noise to be overcome (Kertcher & Zwilling, 2025). It also focuses on specific high-stakes resolution categories like ceasefires, sanctions, and humanitarian aid, rather than general UNSC debating patterns. This work will enable diplomats and political scientists to anticipate rhetorical strategies based on structural position, and recognize how power manifests linguistically in multilateral negotiations.

Our methodology combines five complementary computational methods to test our hypotheses: (1) Stance Detection, categorizing speeches along a five-point scale from "Strong Support" to "Strong Opposition" using rule-based keyword matching combined with vote labels; (2) Rhetorical Strategy Analysis, quantifying ten persuasive techniques through normalized keyword and phrase frequencies; (3) Topic Modeling, applying both Latent Dirichlet Allocation (LDA) and BERTopic to discover latent thematic structures; (4) Framing Analysis, detecting how countries construct identical situations through ten distinct interpretive lenses; and (5) Transformer-Based Embeddings, generating 384-dimensional semantic vectors using SentenceTransformers for similarity analysis and clustering.

2 Related Works

Computational text analysis has been established as a core methodology in political science, with

increasing application to multilateral diplomacy. Laver & Benoit (2018) introduce a language-blind method for estimating policy positions by treating texts as quantitative data, successfully replicating published policy estimates across multiple languages. Grimmer & Stewart (2013) provide a comprehensive guide to automated text analysis, arguing that while these methods reduce the cost of analyzing large text collections, they require careful validation and cannot replace substantive interpretation.

Building on these methodological foundations, international relations scholars have turned to the United Nations as a rich data source for measuring state preferences. Voeten (2012) and Bailey et al. (2015) critique existing UN General Assembly voting measures like dyadic similarity scores, advocating for sophisticated spatial models that separate agenda changes from genuine preference shifts. Bailey et al. (2017) position states along a single dimension reflecting their stance toward the US-led liberal order. Extending beyond voting data, Baturo et al. (2017) introduce the UN General Debate Corpus containing over 7,300 country statements (1970-2014), applying methods like Wordscores to infer policy preferences. Yanmaz (2023) demonstrates that regime type can be predicted from General Debate speeches using supervised learning, achieving 85% accuracy and proving that linguistic patterns encode political characteristics.

Focusing specifically on the Security Council, Schoenfeld et al. (2019) provide the UNSC Debates dataset (1992-2023) with 106,302 speeches to support quantitative analysis. Recent works employ computational methods to analyze how member states and international actors use language in this setting. Johnstone (2005) examines discursive power through legal and normative evolution, demonstrating that deliberation matters in shaping international responses to the use of force, even when material power ultimately drives decisions. Eckhard et al. (2023) apply speaker-topic network analysis to Afghanistan debates (1995-2017), finding that UN bureaucrats exercise autonomous agency by introducing new topics like security reform and elections. Sakamoto (2023) uses word embeddings on 90,000 policy statements (1990-2019) to show that global security discourse remains stratified around terrorism and weapons of mass destruction WMD proliferation, with only limited convergence among P5 members beyond these abstract threats.

More recent work examines emotional and argumentative dimensions. Holtseva (2024) quantifies emotions across 270 meetings in 2023, identifying interest, anger, and fear as dominant affects. Anisimova's ongoing thesis applies Appraisal Theory and argument mining to distinguish factual statements from attitudinal expressions, revealing that diplomatic "facts" are often evaluative reasons supporting particular stances.

However, researchers applying NLP to UN speeches highlight significant challenges. Zaczynska et al. (2024) and Kertcher and Zwilling (2025) found that indirect phrasing, hedging, and strategic vagueness made classification and sentiment analysis difficult, yielding low F1-scores. Liang et al. (2025) introduce UNBench to evaluate LLMs on UNSC tasks, demonstrating that advanced models show promise for geopolitical applications. Despite these advances, the literature predominantly analyzes content, that is, what is discussed, rather than systematically examining how structural power shapes rhetorical mechanics such as certainty, hedging, and framing strategies across member states.

3 Data Collection and Structure

For our experiments, English transcripts of Council meetings (S/PV) are the primary data source. We create a dataset of verbatim meeting records from the United Nations Security Council spanning 75 years, from 1950 to 2025, downloaded from the United Nations Digital Library. The dataset consists of 2069 meeting transcripts for all of the three topics of focus combined. There are 601 records on humanitarian aid, 942 records on sanctions, and the final 526 on ceasefire. For each meeting record (a PDF file), we extract the names and speeches of each country present and save them as a pandas dataframe.

4 Methods and Experiments

4.1 Overview of the Methodology

The resulting framework integrates five complementary analytical components: stance detection, rhetorical strategy analysis, topic modeling, framing analysis, and transformer-based semantic embeddings. These components operate at different linguistic levels: lexical, syntactic-proxy, thematic, and semantic. Together, they enable the extraction of both explicit surface-level signals (e.g., rhetorical devices and framing cues) and latent semantic structure (e.g., contextual similarity between

speeches). This layered design reflects the complexity of diplomatic language, which often relies on indirect phrasing, hedging, and institutional norms rather than explicit position-taking (Kertcher & Zwilling, 2025).

4.2 Stance Detection

Objective: The objective of stance detection is to computationally estimate where each speech lies along a support–opposition continuum with respect to the draft resolution under discussion.

Method: We operationalize stance as a five-point ordinal variable: Strong Support, Weak Support, Neutral, Weak Opposition, and Strong Opposition. Stance is inferred using a rule-based lexical scoring system, designed to capture explicit and implicit evaluative language commonly used in diplomatic discourse. Each speech is tokenized and matched against curated lexicons representing:

- Supportive expressions (e.g., "welcomes," "supports," "endorses"),
- Oppositional expressions (e.g., "rejects," "cannot support"),
- Neutral or hedging expressions (e.g., "notes," "takes into consideration").

For each category, matched terms are aggregated and normalized by speech length to produce comparable scores. These scores are then combined into four derived quantitative measures: Support intensity, Opposition, intensity and Neutrality score. Certainty score, capturing the degree of modal commitment versus hedging.

Experimentation: Rather than optimizing predictive accuracy, experiments analyze stance score distributions across resolution types (ceasefire, sanctions, humanitarian aid) and geopolitical blocs. This allows identification of cases where linguistic stance diverges from formal voting behavior, highlighting the strategic ambiguity characteristic of UNSC rhetoric.

4.3 Rhetorical Strategy Analysis

Objective: This component aims to quantify persuasive rhetorical techniques employed by diplomats to construct legitimacy, justify positions, and appeal to international norms.

Method: We define ten rhetorical strategies grounded in political communication and international relations theory:

- Appeals to authority, Moral framing, Emotional appeals, Urgency framing, Coalition building, Historical precedent, Sovereignty defense, Diplomatic hedging, Assertive language, and Peace emphasis.

Each strategy is operationalized through a custom lexicon, constructed iteratively through corpus inspection and theoretical grounding. For each speech, strategy usage is computed as a normalized frequency score, defined as the proportion of tokens associated with each rhetorical category relative to total speech length.

This normalization ensures comparability across speeches of varying length and speaking style, producing a rhetorical feature vector for each speech.

Experimentation: We analyze rhetorical strategy distributions across resolution types and geopolitical blocs (Western vs. Non-Western). Rather than focusing on absolute frequency, experiments emphasize relative emphasis patterns, identifying rhetorical "fingerprints" that reflect systematic differences in persuasive strategy selection conditioned on issue context.

4.4 Topic Modeling

Objective: Topic modeling is used to uncover latent thematic structure in diplomatic discourse that is not explicitly encoded through keywords or rhetorical markers.

Method: We employ a dual-topic modeling strategy to balance interpretability and semantic expressiveness:

> **Latent Dirichlet Allocation (LDA):** Speeches are represented using a bag-of-words model with standard preprocessing. We train a five-topic LDA model, yielding: Topic–word probability distributions and Per-speech topic mixture vectors.

This probabilistic representation allows direct comparison of thematic emphasis across speeches and resolution types.

> **BERTopic:** To capture contextual semantics beyond lexical overlap, we apply BERTopic, which combines transformer-based sentence embeddings with density-based clustering. This approach identifies semantically coherent topic clusters and supports dynamic topic discovery across heterogeneous speech content.

Experimentation: We compare topic prevalence across resolution categories and time periods. LDA provides interpretable thematic baselines, while BERTopic reveals finer-grained semantic distinctions among speeches that employ similar vocabulary to express different positions.

4.5 Framing Analysis

Objective: Framing analysis examines how identical situations are narratively constructed differently through selective emphasis and interpretation.

Method: We define ten framing lenses: Aggression, Legal violation, Humanitarian crisis, Security threat, Sovereignty, Diplomacy, Self-defense, Non-interference, Justice and accountability, and Emergency.

Frames are detected using a hybrid lexical-intensity approach: Binary indicators capture frame presence and continuous intensity scores measure lexical density associated with each frame.

This design allows both categorical and gradient analysis of framing behavior.

Experimentation: We compare frame usage across resolution types and blocs. Results reveal convergence on humanitarian framing but systematic divergence on sovereignty and accountability frames, illustrating how framing functions as a strategic discursive tool rather than a purely descriptive device.

4.6 Transformer-Based Semantic Embeddings

Objective: Semantic embeddings provide a dense vector representation of speeches that enables similarity analysis without reliance on predefined labels or lexicons.

Method: We encode each speech using SentenceTransformers, producing 384-dimensional contextual embeddings. These embeddings capture semantic relationships informed by large-scale pretrained language models. The embeddings are used for Pairwise cosine similarity analysis, Unsupervised clustering, Zero-shot classification experiments and Dimensionality reduction using UMAP for visualization.

Experimentation: We conduct clustering experiments with embeddings colored by: Meeting topic and Geopolitical bloc. Across experiments, meeting topics consistently dominate the clustering structure, while bloc-level patterns appear as weaker but persistent secondary signals. This suggests that institutional context exerts stronger influence on diplomatic language than national identity alone.

5 Analysis

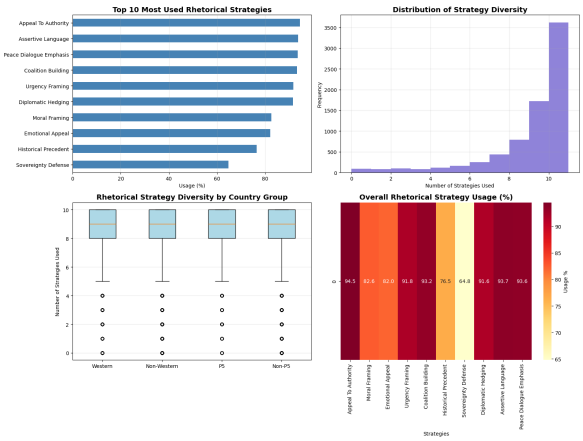

Figure 1: Rhetorical Analysis Results

Figure 1 shows the distribution of rhetorical strategy usage and diversity. Of the ten rhetorical strategies, the most used in the speeches were Appeal to Authority (used in 94.5% of the speeches), Assertive Language (used in 93.7% of the speeches), and Dialogue Emphasis (used in 93.6% of the speeches). The least used strategies were Sovereignty Defense (used in 82.0% of the speeches), Historical Precedent (used in 76.5% of the speeches), and Emotional Appeal (used in 64.8% of the speeches). When differentiated by regional blocs and permanent membership, both Western and Non-Western countries, as well as P5 and non-P5 countries showed a similar rate of rhetorical strategy usage and diversity, with countries usually using eight to ten strategies per speech. This indicates a high rhetorical diversity in the speeches made on the council and the uniformity of usage across countries.

Figure 2 shows the distribution of topics in the speeches by usage (measured by frequency per 100 words) across the three focus areas. We identified 5 topics namely: Sovereignty & non-interference, Humanitarian Crisis & Protection, International Law & Charter Violations, Peaceful Resolution & Dialogue, and Security Threats & Sanctions. The five topics captured distinct rhetorical frames used consistently. Most speeches had high topic-assignment confidence, which shows a clear separation between themes. From this analysis, Sovereignty and Security emerged the most frequently discussed topics across all speeches.

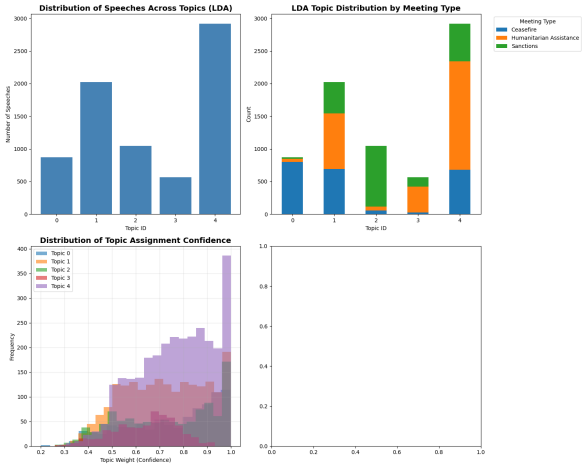

Figure 2: Topic Modeling Analysis Results

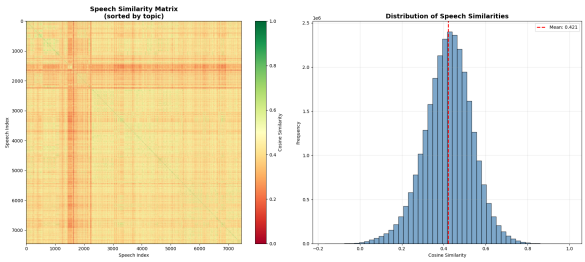

Figure 3: SentenceTransformers Analysis Results

Figure 3 shows the cosine similarity matrix of speeches and their distribution. We observe that speech similarity is moderate across the corpus, with cosine similarity values centered around approximately 0.42. The presence of brighter horizontal and vertical bands indicates that certain issues, such as humanitarian assistance, rely on more standardized and repeatedly used rhetorical templates. These patterns suggest that diplomatic language in the security council is moderately similar across countries and meetings, with recurring phrasing used across institutional context.

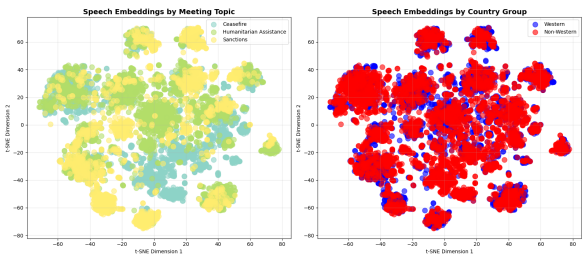

Figure 4: Transformers speech embeddings Analysis Results

Figure 4 shows a two-dimensional projection of sentence level speech embedding. It demonstrates that speech embeddings clearly cluster according to the meeting type, with ceasefire, humanitarian assistance, and sanctions speeches taking up specific areas of the embedding space. This implies that the meeting agenda strongly shapes the semantic content of speeches, independent of the nation in question. While there is some partition between Western and non-Western speeches, this distinction is weaker than clustering by meeting topic, signaling that rhetorical variation is more tightly bound by institutional context than by geopolitical alignment.

6 Evaluations

Given the absence of large-scale human-annotated ground truth for diplomatic discourse phenomena such as stance, framing, and rhetorical strategy, we adopt an evaluation strategy combining internal consistency checks, alignment with external metadata, and qualitative validation. Rather than relying on predictive accuracy, we assess whether each method produces stable, interpretable, and substantively meaningful representations consistent with known institutional and geopolitical patterns.

For stance detection, evaluation emphasizes directional alignment and distributional validity rather than classification accuracy. Distributional validity is assessed by comparing stance intensity distributions across resolution types using non-parametric tests, verifying systematic differences between ceasefire, sanctions, and humanitarian resolutions. Internal coherence is evaluated using pairwise correlations between support, opposition, neutrality, and certainty scores, ensuring theoretically consistent relationships, such as increased hedging correlating with lower certainty.

Rhetorical strategy and framing analysis are evaluated through cross-group differentiation and robustness checks. We test whether normalized strategy and frame frequencies differ across resolution types and geopolitical blocs using group-level mean comparisons and effect sizes. Stability is assessed by recomputing distributions under alternative normalization and pre-processing choices and measuring rank-order consistency. Qualitative inspection of high-scoring examples supports interpretability and theoretical plausibility.

For topic modeling, evaluation follows standard unsupervised metrics and interpretability criteria. LDA models are assessed using topic coherence scores, while BERTopic outputs are evaluated using intra-cluster cosine similarity and inter-cluster sep-

aration. Transformer-based embeddings are evaluated using clustering diagnostics, including silhouette scores and UMAP visual separability, and by comparing clustering strength across metadata labels to test whether embeddings cluster more strongly by meeting topic than by country.

7 Conclusion

In this paper we considered how speech at Security Council meetings is used to justify state rhetoric on the resolutions on ceasefires, sanctions, and humanitarian aid measures. Utilizing a multi-method discourse analysis method that integrates stance detection, rhetorical strategy analysis, framing analysis, topic modeling, and transformer-based semantic embeddings, we examined how countries signal backing, opposition, and lack of clarity through diplomatic language in high impact institutional settings. In every methodological approach, the major factor driving the linguistic variation is meeting topic, not country identity, bloc membership, or vote outcomes. Speeches clustering within resolution type suggest that diplomatic behavior on the Security Council is extremely limited and formulaic. This only marginal variation reflects, to varying degrees, the standardized and highly formal conventions of UN speeches, so speakers must follow universal norms of respect, procedural language and institutional decorum irrespective of geopolitical alignment. Where we might notice some slight differences between Western and non-Western nations, especially regarding framing strategies and rhetorical emphasis, they do not derive from different speech structures, but instead are based on a common institutional template. Both of these findings combined suggest that Security Council statements are less indicators of state behaviour than strategic rhetorical performances molded by institutional norms, audience expectations, and the imperative to maintain negotiation flexibility. This study not only shows that ambiguity, hedging and indirect language are more or less in the middle of linguistic negotiation of power, legitimacy and authority in the Council, but rather it argues that they are features in the making of such negotiating. From a methodological perspective, the project underscores the importance of moving analytical focus away from content extraction for rhetorical profiling. Through adopting framing, certainty, and argumentative structure as an initial focus of study, this work provides a framework for more rigorously and systematically investigating the extent to which institutional background and structural stance influences diplomatic language in multilateral decision-making.

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